

Samanyu Badam

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EDUCATION

Georgia Institute of Technology

Atlanta, GA

Bachelor of Science in Computer Science and Mathematics

May 2028

- Relevant Courses: **Intro to Machine Learning, Data Structures & Algorithms**, Proof-Based Linear Algebra, Applied Combinatorics, Foundations of Mathematical Proofs, Real Analysis 1, Statistics and Applications
- **Honors:** USACO Platinum; AIME Qualifier (4×); AI Safety Initiative Technical Fellow, 2nd Place Citadel Engineering Challenge, 78th (25th in USA) IMC Trading Prosperity 4, 1st Place CalHacks, 1st Place HackGT

SKILLS

Languages: Python, C/C++, Java, TypeScript, SQL, RISC-V (Assembly)

Technologies: PyTorch, NumPy, Numba, Databricks, PySpark, MLflow, Delta Lake, Airflow, GTSAM, nauty, NetworkX, TensorFlow, XGBoost, **ROS2**, MoveIt, OpenCV, NanoOWL, Docker, AWS, Linux, Springboot,

EXPERIENCE

Machine Learning Engineer Intern

May 2026 – Present

Kroger Digital and Technology

Cincinnati, OH

- Trained markdown price-optimization models on **Databricks** to recover maximum margin on perishable inventory before expiry, improving projected **sell-through by 11%** over the rule-based baseline across **6** categories.
- Built an **Apache Airflow**-orchestrated data layer (**PySpark**, **Delta Lake**) ingesting daily movement, quantity-on-display, and balance-on-hand from **180** stores, cutting model-experiment data prep by **70%**.
- Designed the model evaluation framework—**slice-based eval** across **36** category/region segments plus behavioral **invariance tests** (markdown monotonic in time-to-expiry)—surfacing **3** failure modes and guiding data curation.

Undergraduate Researcher

Jan. 2026 – Present

Advisor: Frank Dellaert School of Interactive Computing, Georgia Tech

Atlanta, GA

- Designed a **spectral-residual graph transformer** trained with **reinforcement learning** to order SLAM factor graphs, beating METIS on **90% of 404** held-out graphs with **24%** lower mean max-clique size.
- Built a neural-partition + local min-fill refinement decoder within **7%** of min-fill quality at **2–5×** lower runtime; diagnosed cross-size generalization failure via **gradient-direction analysis**.
- Evaluated across **9 SLAM benchmarks** (800–5,000 nodes); discovered that randomized tie-breaking in min-fill yields lower treewidth than the deterministic version—a zero-cost improvement to existing practice.

AI Researcher

May 2025 – August 2025

Origami Robotics (YC-S25)

Atlanta, GA

- Designed AI task-generation layer for modular reconfigurable robots, mapping natural-language intent to executable **behavior-tree XML** pipelines with typed I/O contracts, ensuring faithful translation from human specification to robot behavior.
- Deployed **perception-to-planning integration** (NanoOWL + MoveIt) on **NVIDIA Jetson** for real-time object detection, frame-transform calibration, and collision-aware trajectory planning in safety-critical manipulation.
- Built **graph-based orchestration** connecting behavior-tree structure to ROS2 skill nodes, enabling the same kinematic stack to reconfigure between pick-and-place, CNC, and 3D-printing modes.

PROJECTS & PUBLICATIONS

Graph-Structural Signatures of Deceptive Circuits in LLMs | *Python, SAELens, NetworkX*

May 2026

- Built the first application of **treewidth analysis** to SAE feature circuits, testing whether graph topology discriminates honest from deception-adjacent behavior in GPT-2 across **150 prompts** and **183 co-activation graphs**.
- Achieved **80% classification accuracy** (AUC 0.851) with 18 structural measures; treewidth alone captures 78.7%, with honest computation more entangled than confabulation (Cohen's $d = -1.24$, $p < 10^{-8}$).
- Implemented Numba-accelerated treewidth scoring, SAELens/TransformerLens feature extraction, and statistical pipeline (Mann-Whitney U, Bonferroni, RF ablation); **open-sourced** with reproducible configs.

Vertex-Minor Ramsey Number $R_{vm}(4)$ | *C, nauty, Python*

March 2026 – April 2026

- Proved $R_{vm}(4) > 10$ by exhaustively checking all **12,005,168** non-isomorphic 10-vertex graphs for containment of I_4 as a vertex-minor via BFS over local equivalence classes, completing in **11 seconds**.
- Implemented single-XOR local complementation via **precomputed toggle masks**, generation-counter hash sets for $O(1)$ clearing, and bitwise independence-set pruning; found exactly **6 obstruction graphs** sharing one LC class of 8,712 members.
- Built **115-test** verification suite confirming max independence number of 3 across the obstruction orbit; **open-sourced** with reproducible pipeline via nauty. Motivated by arXiv:2602.09049.